2505Av2m2 math

1. If $9 + x = 6$, what is the value of $-18 - 2x$?

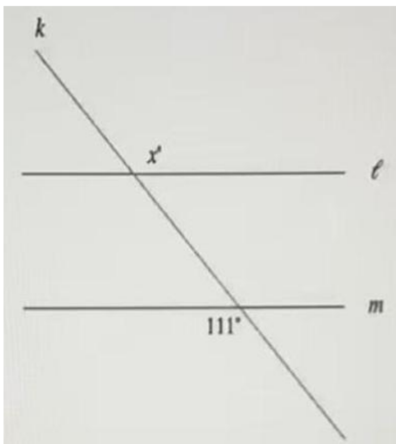
   A. -12

   B. -2

   C. 6

   D. 24

2. In the figure, line l is parallel to line m. What is the value of x?

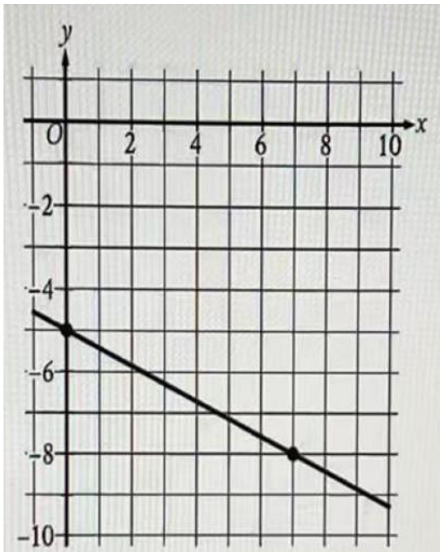
   A. 31

   B. 69

   C. 111

   D. 131

3. The graph of the linear function f is shown, where $y = f(x)$. Which equation defines f?



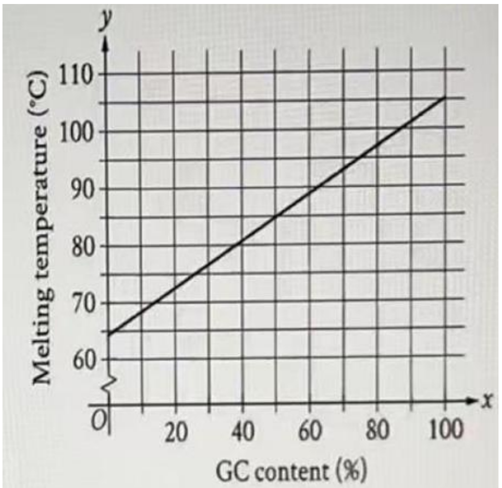
A. $f(x) = \frac{3}{7}x - 5$

B. $f(x) = \frac{7}{3}x - 5$

C. $f(x) = -\frac{3}{7}x - 5$

D. $f(x) = -\frac{7}{3}x - 5$

4. The function t gives the estimated melting temperature, in degrees Celsius, of a DNA molecule as a function of the percent of guanine-cytosine (GC) content, x, in the DNA molecule. The graph of $y = t(x)$ is shown. According to the graph, which of the following statements is NOT true?



A. For each increase of x by 1, $t(x)$ increases by approximately $\frac{2}{5}$.

B. A DNA molecule with a GC content of 0% has an estimated melting temperature between 60°C and 65°C.

C. A DNA molecule with a GC content of 95% has an estimated melting temperature between 100°C and 105°C.

D. A DNA molecule with a GC content of 98% has an estimated melting temperature between 80°C and 90°C.

5. In the xy-plane, line s is parallel to line t. Line s is defined by $y + 16 = 7x$. What is the slope of line t?

A. -16

B. -7

C. 7

D. 16

6. Scientists collected fallen acorns that each housed a colony of the ant species *P.ohioensis* and analyzed each colony's structure. For any of these colonies, if the colony has x worker ants, the equation $y = 0.67x + 2.6$, where $20 \leq x \leq 110$, gives the predicted number of larvae, y, in the colony. If one of these colonies has 35 worker ants, which of the following is the closest to the predicted number of larvae in the colony?

   A. 29

   B. 42

   C. 54

   D. 128

7. For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

$y > 5x - 4$

A.

x	y
4	11
6	21
9	36

B.

x	y
4	16
6	26
9	41

C.

x	y
4	21
6	31
9	46

D.

x	y
4	21
6	21
9	41

8. The given equation relates the variables, P, N, and C. Which expression represents the value of C for distinct positive values of P and N?

   $P = N(34 - C)$

   A. $C + \frac{P}{N}$

   B. $34 + \frac{P}{N}$

   C. $34 - \frac{P}{C}$

   D. $34 - \frac{P}{N}$

9. What is the sum of the solutions of the given equation?

   $x^2 - 15x + 12 = 0$

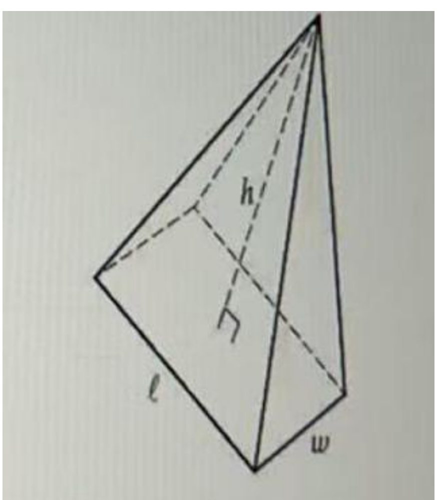
   A. 15

   B. 12

   C. 3

   D. 0

10. The figure shown is a right rectangular pyramid, where $t = 18$ units, $w = 9$ units, and $h = 12$ units. What is the surface area, in square units, of the pyramid? (Note: Round the answer to the nearest tenth.)



Note: Figure not drawn to scale.

11. The function $f(x) = \frac{1}{7}(x - 6)^2 + 4$ gives a toy car's height above the ground $f(x)$, in inches, x seconds after it started moving on an elevated track, where $0 \le x \le 10$. Which of the following is the best interpretation of the vertex of the graph of $y = f(x)$ in the xy-plane?

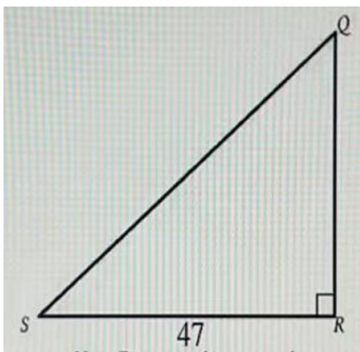
A. The toy car's minimum height was 4 inches above the ground.

B. The toy car's minimum height was 6 inches above the ground.

C. The toy car's height was 4 inches above the ground when it started moving.

D. The toy car's height was 6 inches above the ground when it started moving.

12. In the triangle QRS shown, $QR < RS$. Which expression represents the length of QS?



Note: Figure not drawn to scale.

A. $47cosQ$

B. $47sinQ$

C. $\frac{47}{cosQ}$

D. $\frac{47}{sinQ}$

13. A researcher surveyed undergraduate students, graduate students, and postdoctoral students. The number of undergraduate students surveyed was 7,150% of the number of postdoctoral students surveyed, and the number of graduate students surveyed was 45% of the number of undergraduate students surveyed. If there were 6,435 graduate students surveyed, what was the sum of the number of undergraduate students and postdoctoral students surveyed?

14. In the given system of equations, t is a constant. If the system has no solution, what is the value of t?

$16x - 20y = 6y + 8$

$ty = 8x + \frac{1}{5}$

15. The function f is defined by the given equation. If $g(x) = f(x + 3)$, which of the following equations defines the function g?

$f(x) = 7(4)^x$

A. $g(x) = 21(4)^x$

B. $g(x) = 448(4)^x$

C. $g(x) = 21(12)^x$

D. $g(x) = 343(64)^x$

16. The bone mineral density of a certain person's thoracic spine was estimated to be 0.812 grams per square centimeter. What is this estimated bone mineral density in grams per **square millimeter**? (1 centimeter = 10 millimeters)

   A. 81.2

   B. 8.12

   C. 0.0812

   D. 0.00812

17. Which expression is a factor of $49p^{19} - 121p^{17}$?

   A. $7p + 11$

   B. $49p^2 + 121$

   C. $11p - 7$

   D. $-72p^{40}$

18. In the given equation, k is a positive constant. The product of the solutions to the equation is 72. What is the value of k?

$$\frac{5}{9}(5x + 9)\left(x + \sqrt{5k + 9}\right)\left(x - \sqrt{5k + 9}\right) = 0$$

19. A right rectangular prism has a base area of 56t square centimeters (cm^2). The length of the base of the rectangular prism is $\frac{14}{3}$ cm, and the height of the rectangular prism is 3 cm. Which expression represents the surface area, in cm^2, of the right rectangular prism?

A. $112t + 56$

B. $184t + 28$

C. $1{,}680t + 28$

D. $168t$

20. In the xy-plane, a unit circle with center at the origin O contains point A with coordinate (1,0) and point B with coordinate $(\frac{5}{\sqrt{34}}, -\frac{3}{\sqrt{34}})$. If the measure of angle AOB is p radians, what is the value of $\frac{\cos(p)}{\sin(p)}$?

21. The function h is defined by $h(x) = -\sqrt{x^2 + bx + c}$, where b and c are constants. In the xy-plane, the graph of $y = h(x)$ contains $(2,0)$ and $(0, -\sqrt{334})$. If $h(m) = 0$, what is the greatest possible value of m?

22. A computer repair specialist charges $160 for the first two hours of repair plus an hourly fee for each additional hour. The total cost for 6 hours of repair is $360. Which function f gives the total cost, in dollars, for x hours of repair, where $x \geq 2$?

A. $f(x) = 50x + 60$

B. $f(x) = 50x + 160$

C. $f(x) = 60x$

D. $f(x) = 60x + 160$